Graded Project on Orchestration and Scaling

Jenkins Access Details
URL: https://jenkinsacademics.herovired.com/
Username: herovired
Password: herovired
Project Repository

Main Repository: https://github.com/UnpredictablePrashant/StreamingApp.git

Instructions to Sync Fork with Upstream: Pull updates from original GitHub repository into a fork

Project Workflow

Step 1: Version Control with Git
Fork the main repository into your GitHub account.
Maintain your version by syncing updates from the main repository as needed.

Step 2: Prepare the MERN Application
Containerize the Application
Create Dockerfiles for each component (Frontend and Backend).Push Docker Images to Amazon ECR
Build Docker images for both components.Create individual ECR repositories.
Push the images to their respective ECR repositories.

Step 3: AWS Environment Setup
Install and Configure AWS CLI
Set up AWS CLI with your access credentials.

Step 4: Continuous Integration (CI) using Jenkins
Set Up Jenkins
Install Jenkins on an AWS EC2 instance.
Configure necessary plugins and credentials.Create Jenkins Pipelines
Build and push Docker images to ECR using Jenkins jobs.
Trigger builds automatically on new commits from the Git repository.

Step 5: Kubernetes Deployment (EKS)
Create an EKS Cluster
Use eksctl or similar tools to set up Amazon EKS.Deploy via Helm

Package and deploy the MERN stack using Helm charts.

Step 6: Monitoring and Logging
Set Up Monitoring
Use Amazon CloudWatch for metrics and alarms.Configure Logging
Centralize application logs using CloudWatch Logs or similar tools.

Step 7: Documentation
Document the complete system architecture and deployment process.
Include diagrams, configuration steps, and scripts.
Upload the documentation to your GitHub repository.

Step 8: Final Validation
Verify that both frontend and backend are functional and accessible.
Ensure the application meets deployment and scaling requirements.

Bonus: Step 9 — ChatOps Integration
Create SNS Topics
Use SNS topics for various events (e.g., deployment success/failure).Integrate with Messaging Platform
Connect SNS with Slack, Microsoft Teams, or Telegram for real-time alerts.
Submission Instructions
Ensure your project is complete and tested.
Push all code and documentation to your GitHub repository.
Share your repository link in a text, Word, or PDF file.
Upload the file containing the link via Vlearn.

Practice Notes

1 . Fork the Repository

Go to:

👉 https://github.com/UnpredictablePrashant/StreamingApp.git

Click **Fork** → It created:

https://github.com/rajkumarsaw22/StreamingApp.git

2. Clone Fork

git clone https://github.com/rajkumarsaw22/StreamingApp.git

cd StreamingApp

---

3. Added Upstream Repository

This connects your fork with the original repo.

git remote add upstream https://github.com/UnpredictablePrashant/StreamingApp.git

Check remotes:

git remote -v

You should see:

origin → your fork
upstream → original repo

```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git remote -v
origin  git@github.com:rajkumarsaw22/StreamingApp.git (fetch)
origin  git@github.com:rajkumarsaw22/StreamingApp.git (push)
```

```
upstream	https://github.com/UnpredictablePrashant/StreamingApp.git (fetch)
upstream	https://github.com/UnpredictablePrashant/StreamingApp.git (push)
```

4. Sync Fork with Upstream

Whenever the original repo updates, you sync like this:

```
git fetch upstream
git checkout main
git merge upstream/main
git push origin main
```

OR better (clean way):

```
git pull upstream main
git push origin main
```

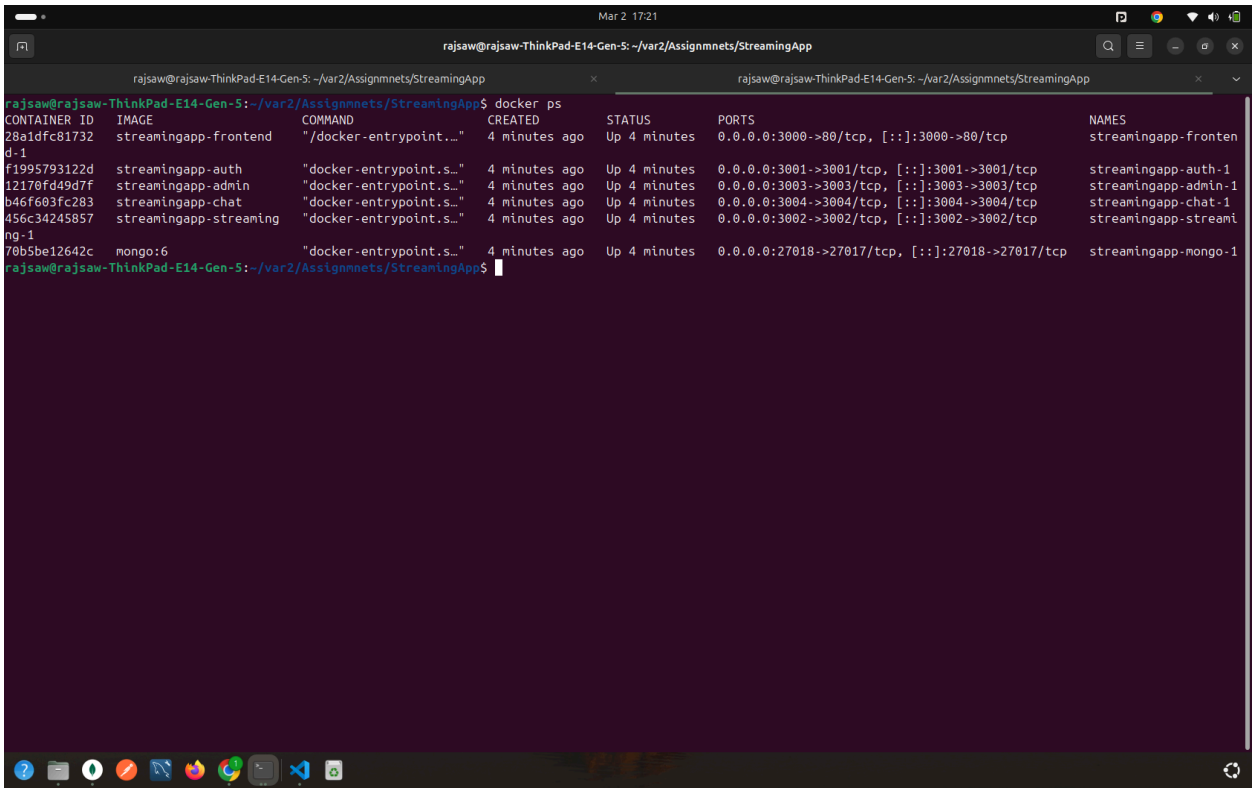
```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git pull
upstream main
From https://github.com/UnpredictablePrashant/StreamingApp
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> upstream/main
Already up to date.
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git push
origin main
Everything up-to-date
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
```

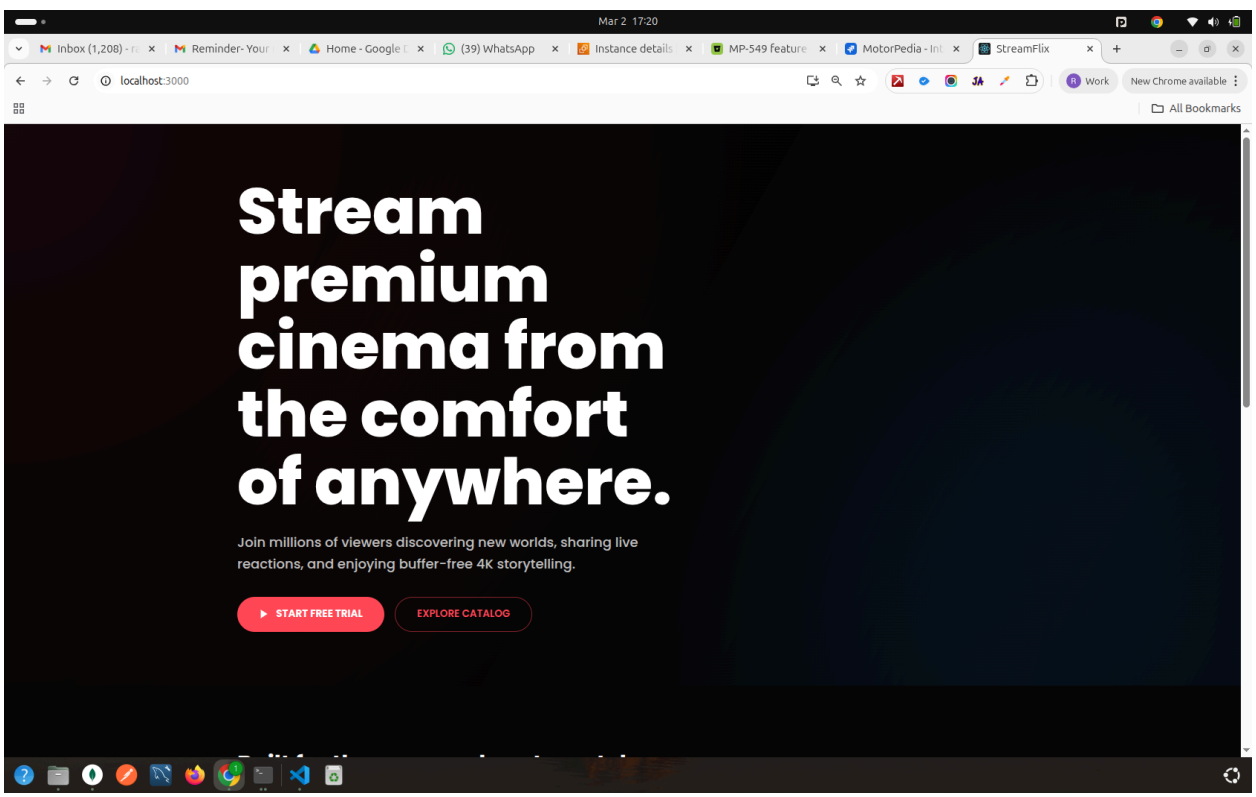
```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git remote add upstream https://github.com/UnpredictablePrashant/StreamingAp
p.git
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git remote -v
origin  git@github.com:rajkumarsaw22/StreamingApp.git (fetch)
origin  git@github.com:rajkumarsaw22/StreamingApp.git (push)
upstream        https://github.com/UnpredictablePrashant/StreamingApp.git (fetch)
upstream        https://github.com/UnpredictablePrashant/StreamingApp.git (push)
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git remote -v
origin  git@github.com:rajkumarsaw22/StreamingApp.git (fetch)
origin  git@github.com:rajkumarsaw22/StreamingApp.git (push)
upstream        https://github.com/UnpredictablePrashant/StreamingApp.git (fetch)
upstream        https://github.com/UnpredictablePrashant/StreamingApp.git (push)
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git pull upstream main
From https://github.com/UnpredictablePrashant/StreamingApp
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> upstream/main
Already up to date.
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ git push origin main
Everything up-to-date
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
```

Git Repo : https://github.com/rajkumarsaw22/StreamingApp

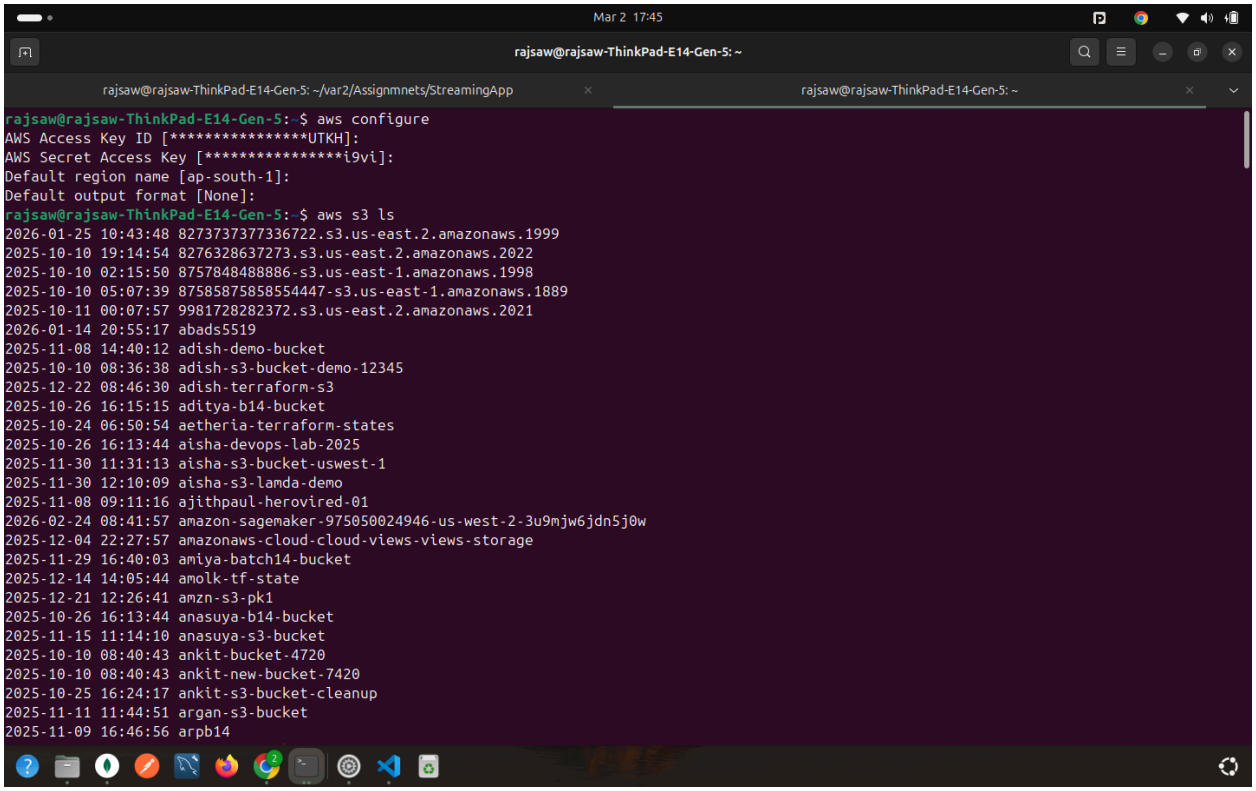
- Created .env file in all services and frontend folder
- Cross checked all dockerfiles
- Run the docker-compose.yml file
- docker compose up --build

Checked its working locally, Once It will work locally, will push images to ECR


Mar 2 17:21
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
28a1dfc81732 streamingapp-frontend "/docker-entrypoint.…" 4 minutes ago Up 4 minutes 0.0.0.0:3000->80/tcp, [::]:3000->80/tcp streamingapp-frontend-1
f1995793122d streamingapp-auth "docker-entrypoint.s…" 4 minutes ago Up 4 minutes 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp streamingapp-auth-1
12170fd49d7f streamingapp-admin "docker-entrypoint.s…" 4 minutes ago Up 4 minutes 0.0.0.0:3003->3003/tcp, [::]:3003->3003/tcp streamingapp-admin-1
b46f603fc283 streamingapp-chat "docker-entrypoint.s…" 4 minutes ago Up 4 minutes 0.0.0.0:3004->3004/tcp, [::]:3004->3004/tcp streamingapp-chat-1
456c34245857 streamingapp-streaming "docker-entrypoint.s…" 4 minutes ago Up 4 minutes 0.0.0.0:3002->3002/tcp, [::]:3002->3002/tcp streamingapp-streaming-1
70b5be12642c mongo:6 "docker-entrypoint.s…" 4 minutes ago Up 4 minutes 0.0.0.0:27018->27017/tcp, [::]:27018->27017/tcp streamingapp-mongo-1
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp$


Mar 2 17:20
Inbox (1,208) - r
Reminder- Your
Home - Google
(39) WhatsApp
Instance details
MP-549 feature
MotorPedia - Int
StreamFlix
localhost:3000
Work
New Chrome available
All Bookmarks
Stream premium cinema from the comfort of anywhere.
Join millions of viewers discovering new worlds, sharing live reactions, and enjoying buffer-free 4K storytelling.
START FREE TRIAL
EXPLORE CATALOG

AWS CLI Login



Build and push Docker images to ECR - First Time manually -

```
mp-services-admin-api:latest              447ea4ac2b25        136MB            0B
mp-services-adminapi:latest               bada384f8f73        135MB            0B    U
mp-services-central:latest                3d9626193e06        135MB            0B
mp-services-core-services:latest          51653dd6b471        928MB            0B    U
mp-services-core:latest                   cb406c0ce980        928MB            0B
mp-services-dc-api:latest                 8f8d21d8374f        136MB            0B
mp-services-dealerauth:latest             3d1ca0d1d733        222MB            0B    U
mp-services-gateway-manager:latest        5a1a998b1998        135MB            0B    U
mp-services-inventory-admin-api:latest    7da8ddd266b4        136MB            0B
mp-services-inventory-dealer-api:latest   02401f789cdc        136MB            0B
mp-services-notification:latest           20a856999dbe        136MB            0B    U
mp-services-pdf:latest                    7e14b4232af4        918MB            0B
mp-services-sfa:latest                    0879c2eb0f3a        137MB            0B    U
streamingapp-admin:latest                 57be5bdfb940        169MB            0B    U
streamingapp-auth:latest                  971de6dae2aa        158MB            0B    U
streamingapp-chat:latest                  85fac02c6784        152MB            0B    U
streamingapp-frontend:latest              cfa4d718ac1b       51.8MB            0B    U
streamingapp-streaming:latest             a9dbdf8bf374        185MB            0B    U
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ aws ecr get-login-password --region us-west-1 | docker login --username AWS --password-stdin 975050024946.dkr.ecr.us-west-1.amazon
aws.com

WARNING! Your credentials are stored unencrypted in '/home/rajsaw/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker tag streamingapp-admin:latest 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-admin:latest
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker push 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-admin:latest
The push refers to repository [975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-admin]
3446dc7beb1b: Pushed
29c977b4d1fe: Pushed
88410ac53df8: Pushed
3a49af32a7b8: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushed
08000c18d16d: Pushed
latest: digest: sha256:b5482f8b2cd3970cf5622684a469e673dc95ef4a5b9b167919338ea3e91cf7c1 size: 1995
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker images
```

Mar 3 17:56

rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~

```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ aws ecr get-login-password --region us-west-1 | docker login --username AWS --password-stdin 975050024946.dkr.ecr.us-west-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/rajsaw/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker tag streamingapp-auth:latest 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-auth:latest
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker push 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-auth:latest
The push refers to repository [975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-auth]
c5203c3e62e9: Pushed
781895dac41: Pushed
e1513b1edd06: Pushed
787399bfaf5c: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushed
08000c18d16d: Pushed
latest: digest: sha256:2ae007345cc3cdb49f2462e7ece9f6c306df608922330fd40176f5e0d1204978 size: 1994
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker tag streamingapp-chat:latest 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-chat:latest
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker push 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-chat:latest
The push refers to repository [975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-chat]
b7b8824d89c9: Pushed
2c75252906ab: Pushed
352fd2740cf4: Pushed
7d8ac27b0ccb: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushed
08000c18d16d: Pushed
latest: digest: sha256:e2b95afabb058798558600fb11064f9a838f7769f5e4810976c4fd90817df6b4 size: 1994
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker tag streamingapp-frontend:latest 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-frontend:latest
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker push 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-frontend:latest
The push refers to repository [975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-frontend]
4e02a6de90cd: Pushed
0d853d50b128: Pushed
947e005a4ac7: Pushed
811a4dbbf4a5: Pushed
b8d7d1d22634: Pushed
e244aa659f61: Pushed
c56f134d3805: Pushed
d71eae0084c1: Pushed
08000c18d16d: Pushed
latest: digest: sha256:b16f79ad4faeb59947dba27de2f09a16406639ae01cbd1979cd2bcdcfee448c3 size: 2199
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker tag streamingapp-streaming:latest 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-streaming:latest
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ docker push 975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-streaming:latest
The push refers to repository [975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-streaming]
be518378f5a5: Pushed
a6fdaa17c3f0: Pushed
3b5426dde43a: Pushed
1c2023365054: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushed
08000c18d16d: Pushed
latest: digest: sha256:752ab110d1978776e81065bf5acaec1d011d1fc3fb54e5531a794c5259fda42a size: 1998
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$
```

Mar 3 17:45

MotorPedia Mail | (23) WhatsApp | Reminder- Your repayme | Home - Google Drive | Graded Project on Orche | Elastic Container Registr

us-west-1.console.aws.amazon.com/ecr/private-registry/repositories?region=us-west-1

Rate your experience with this Elastic Container Registry console.

Amazon ECR > Private registry > Repositories

Amazon Elastic Container Service

- Private registry
  - Repositories
  - Features & Settings
- Public registry
  - Repositories
  - Settings

ECR public gallery
Amazon ECS
Amazon EKS

Getting started
Documentation

Successfully created private repository, rajsaw/streamingapp-streaming

Private repositories (19)

View push commands | Delete | Actions | Create repository

rajsaw

Repository name	URI	Created at	Tag immutability	Encryption type
rajsaw/streamingapp-admin	975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-admin	March 03, 2026, 00:23:20 (UTC+05.5)	Mutable	AES-256
rajsaw/streamingapp-auth	975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-auth	March 03, 2026, 17:41:26 (UTC+05.5)	Mutable	AES-256
rajsaw/streamingapp-chat	975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-chat	March 03, 2026, 17:43:56 (UTC+05.5)	Mutable	AES-256
rajsaw/streamingapp-frontend	975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-frontend	March 03, 2026, 17:44:49 (UTC+05.5)	Mutable	AES-256
rajsaw/streamingapp-streaming	975050024946.dkr.ecr.us-west-1.amazonaws.com/rajsaw/streamingapp-streaming	March 03, 2026, 17:45:25 (UTC+05.5)	Mutable	AES-256

CloudShell | Feedback | Console Mobile App

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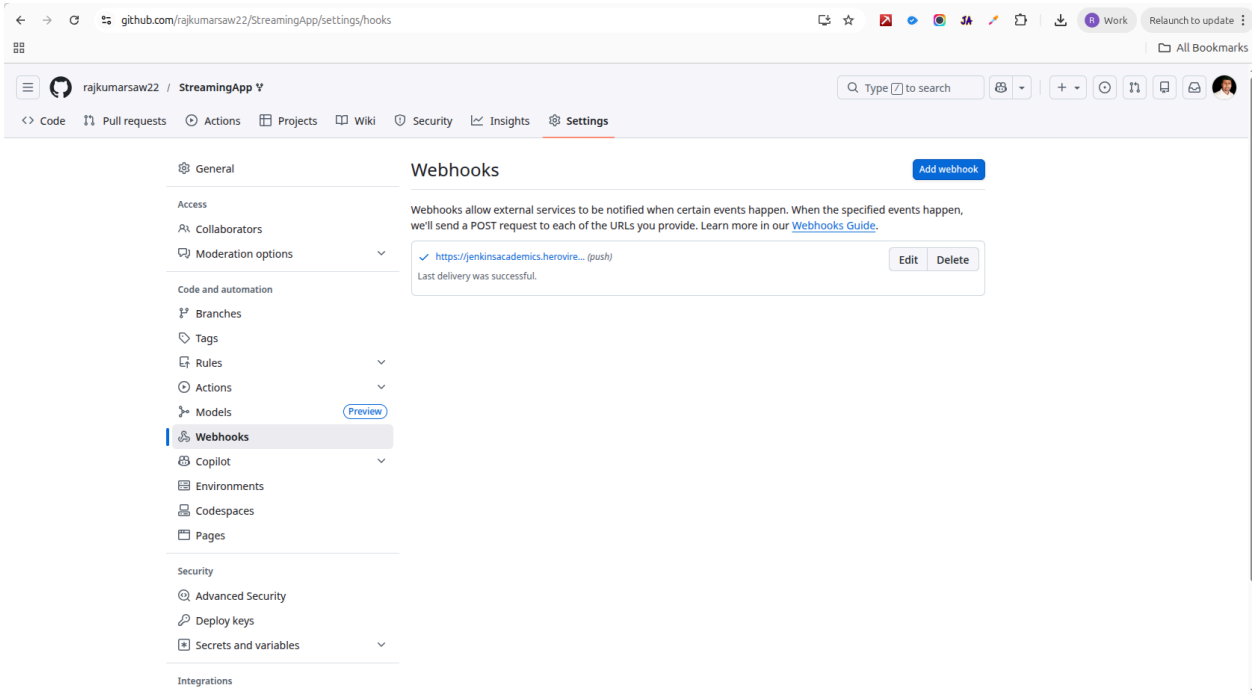
Webhook added in Github



Jenkins / RajSaw-StreamingApp-Mic... / Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

General

Enabled

Description

Plain text Preview

- [ ] Discard old builds
- [ ] Do not allow concurrent builds
- [ ] Do not allow the pipeline to resume if the controller restarts
- [ ] GitHub project
- [ ] Delivery Pipeline configuration
- [ ] Pipeline speed/durability override
- [ ] Preserve stashes from completed builds
- [ ] This project is parameterized
- [ ] Throttle builds

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- [ ] Build after other projects are built
- [ ] Build periodically
- [ ] GitHub hook trigger for GITScm polling
- [ ] Poll SCM
- [ ] Trigger builds remotely (e.g., from scripts)

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/rajkumarsaw22/StreamingApp.git

Credentials

- none -

+ Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main

+ Add Branch

Repository browser

(Auto)

Additional Behaviours

+ Add

Script Path

Jenkinsfile

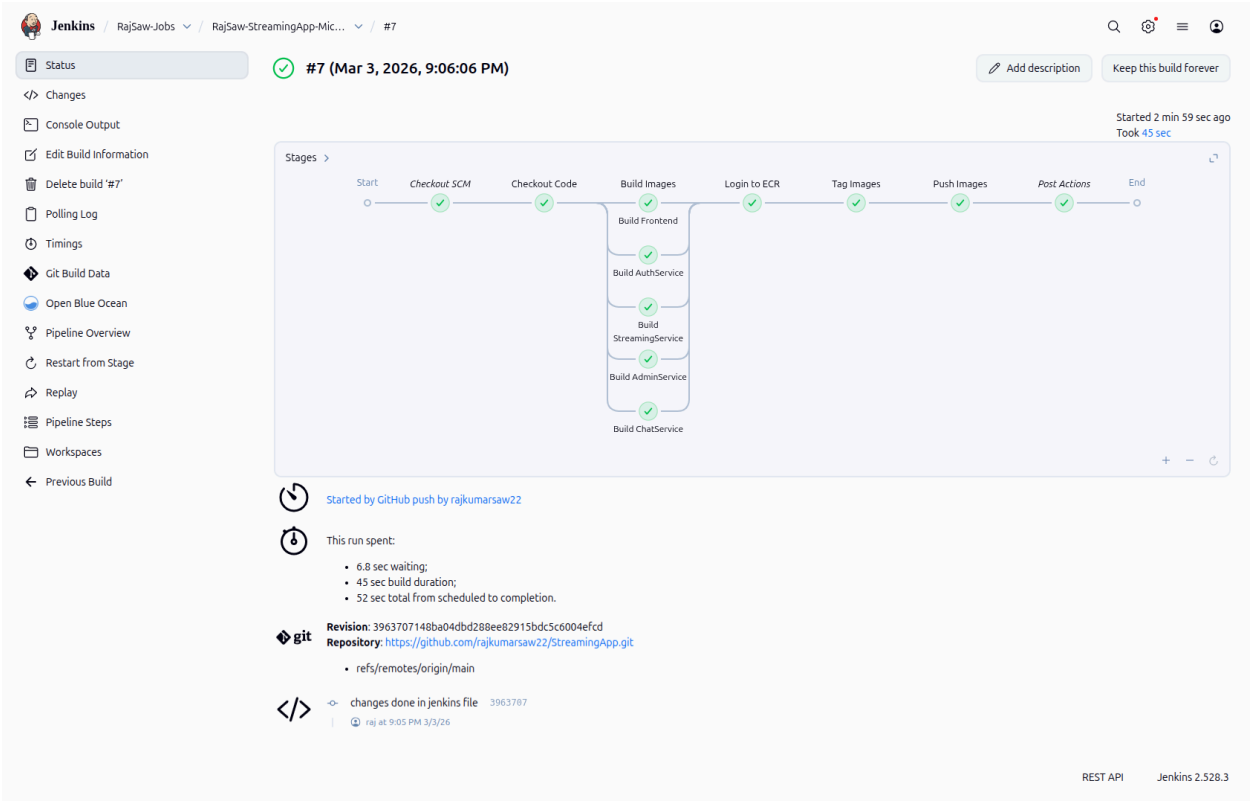
- [x] Lightweight checkout

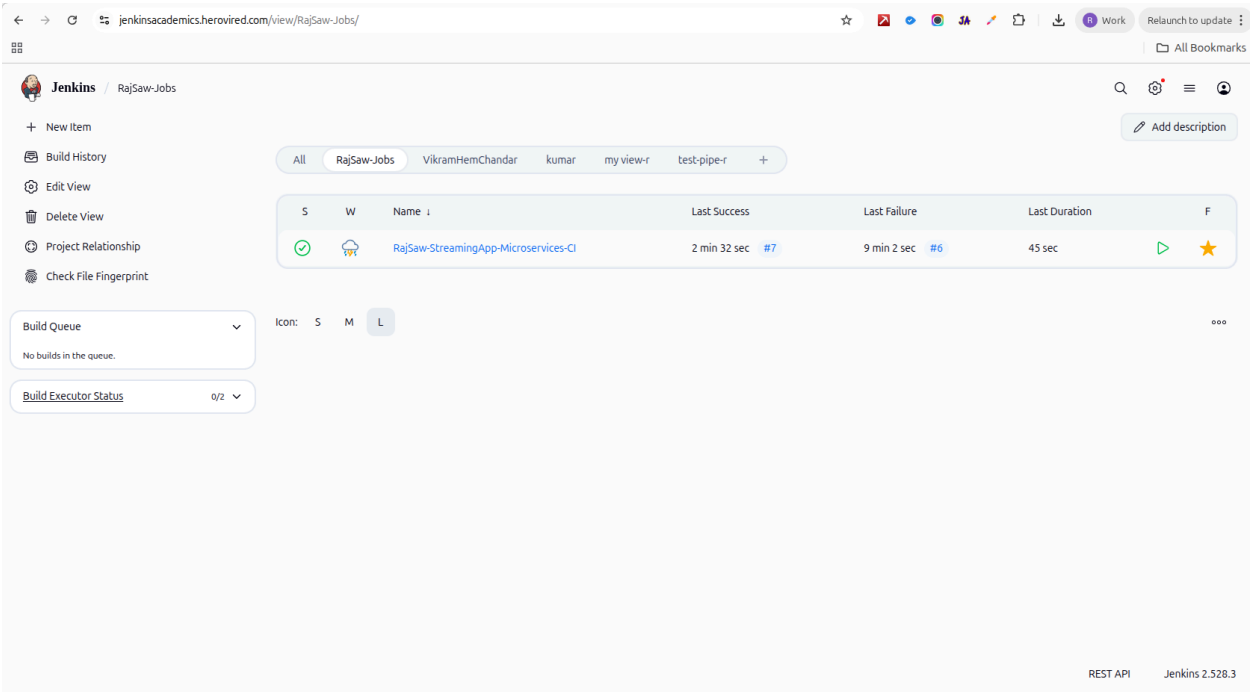
Pipeline Syntax

Advanced

Advanced

Save Apply

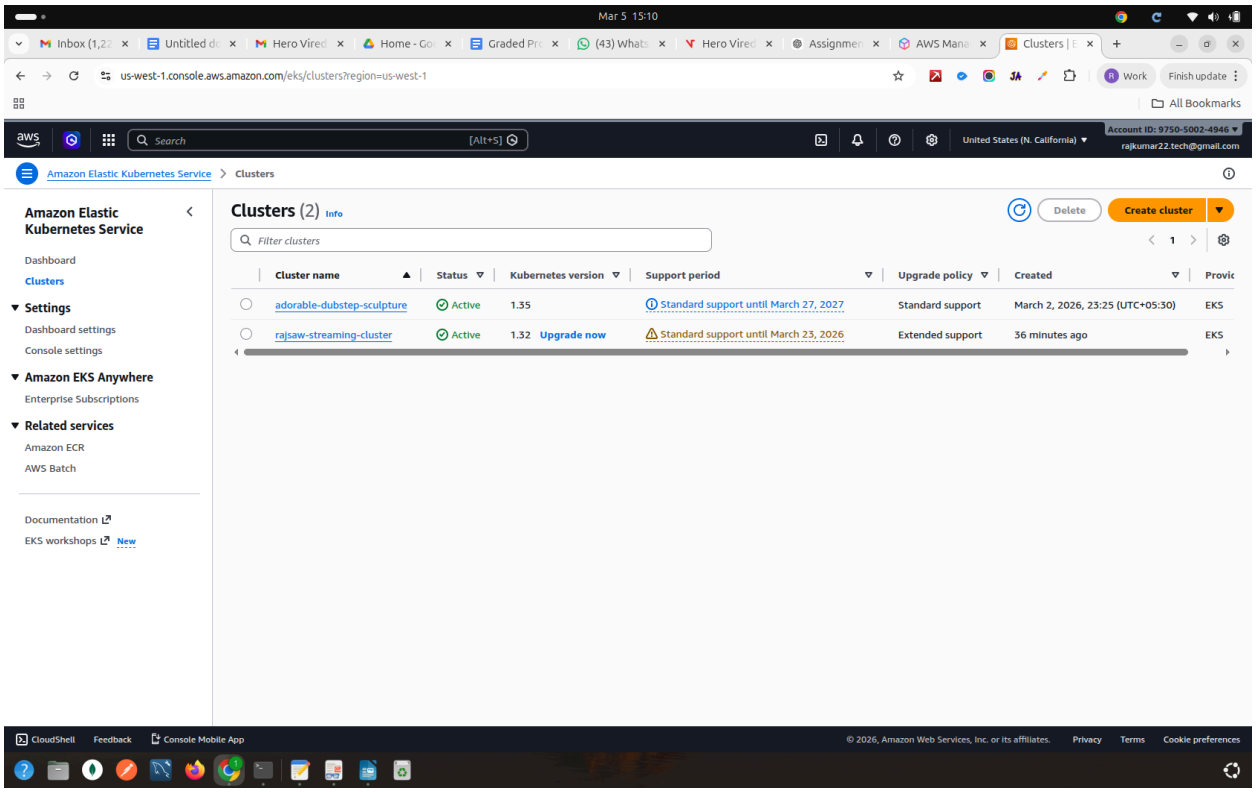

Jenkins / RajSaw-Jobs / RajSaw-StreamingApp-Mic... / #7
Status
Changes
Console Output
Edit Build Information
Delete build '#7'
Polling Log
Timings
Git Build Data
Open Blue Ocean
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#7 (Mar 3, 2026, 9:06:06 PM)
Add description
Keep this build forever
Started 2 min 59 sec ago
Took 45 sec
Stages
Start
Checkout SCM
Checkout Code
Build Images
Login to ECR
Tag Images
Push Images
Post Actions
End
Build Frontend
Build AuthService
Build StreamingService
Build AdminService
Build ChatService
Started by GitHub push by rajkumarsaw22
This run spent:
6.8 sec waiting;
45 sec build duration;
52 sec total from scheduled to completion.
Revision: 3963707148ba04dbd288ee82915bdc5c6004efcd
Repository: https://github.com/rajkumarsaw22/StreamingApp.git
refs/remotes/origin/main
changes done in jenkins file 3963707
raj at 9:05 PM 3/3/26
REST API
Jenkins 2.528.3


jenkinsacademics.herovired.com/view/RajSaw-Jobs/
Work
Relaunch to update
All Bookmarks
Jenkins / RajSaw-Jobs
New Item
Build History
Edit View
Delete View
Project Relationship
Check File Fingerprint
Add description
All
RajSaw-Jobs
VikramHemChandar
kumar
my view-r
test-pipe-r
S
W
Name
Last Success
Last Failure
Last Duration
F
RajSaw-StreamingApp-Microservices-CI
2 min 32 sec #7
9 min 2 sec #6
45 sec
Build Queue
No builds in the queue.
Build Executor Status 0/2
Icon: S M L
REST API
Jenkins 2.528.3

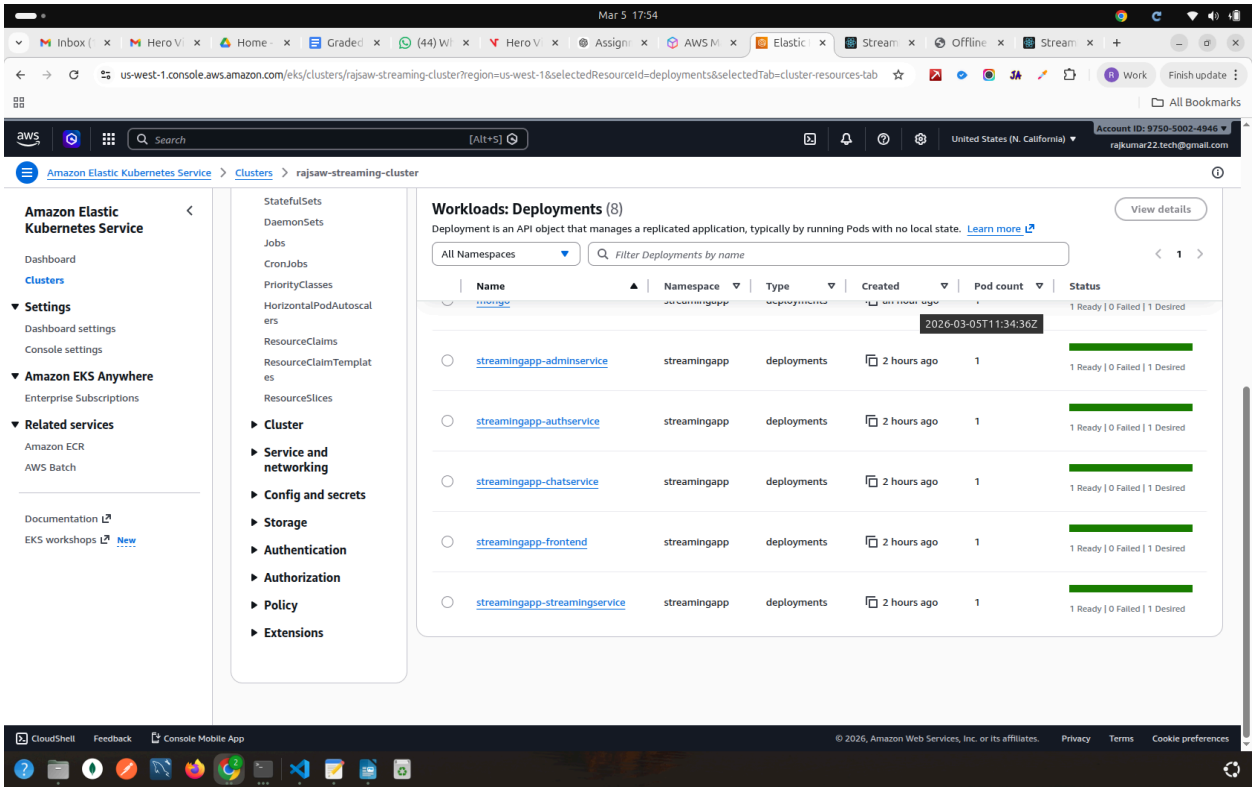
EKS SETUP -

```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ eksctl version
0.221.0
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ eksctl create cluster \
--name rajsaw-streaming-cluster \
--region us-west-1 \
--nodegroup-name streaming-nodes \
--node-type t3.medium \
--nodes 2
2026-03-05 14:34:28 [ℹ]  eksctl version 0.221.0
2026-03-05 14:34:28 [ℹ]  using region us-west-1
2026-03-05 14:34:30 [ℹ]  setting availability zones to [us-west-1a us-west-1c]
2026-03-05 14:34:30 [ℹ]  subnets for us-west-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-03-05 14:34:30 [ℹ]  subnets for us-west-1c - public:192.168.32.0/19 private:192.168.96.0/19
2026-03-05 14:34:30 [ℹ]  nodegroup "streaming-nodes" will use "" [AmazonLinux2023/1.32]
2026-03-05 14:34:30 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no
longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.i
o/usage/auto-mode/
2026-03-05 14:34:30 [ℹ]  using Kubernetes version 1.32
2026-03-05 14:34:30 [ℹ]  creating EKS cluster "rajsaw-streaming-cluster" in "us-west-1" region with managed nodes
2026-03-05 14:34:30 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-03-05 14:34:30 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-west-1 --cluster=rajsaw-streaming-
cluster'
2026-03-05 14:34:30 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "rajsaw-streaming-cluster" in "us-wes
t-1"
2026-03-05 14:34:30 [ℹ]  CloudWatch logging will not be enabled for cluster "rajsaw-streaming-cluster" in "us-west-1"
2026-03-05 14:34:30 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-west-1 --clus
ter=rajsaw-streaming-cluster'
2026-03-05 14:34:30 [ℹ]  default addons kube-proxy, coredns, metrics-server, vpc-cni were not specified, will install them as EKS addons
2026-03-05 14:34:30 [ℹ]
2 sequential tasks: { create cluster control plane "rajsaw-streaming-cluster",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "streaming-nodes",
    }
}
2026-03-05 14:34:30 [ℹ]  building cluster stack "eksctl-rajsaw-streaming-cluster-cluster"
2026-03-05 14:34:32 [ℹ]  deploying stack "eksctl-rajsaw-streaming-cluster-cluster"
```

- Install helm



Deployment Done



```
Mar 5 17:56
rajsaw@rajsaw-ThinkPad-E14-Gen-5: ~/var2/Assignmnets/StreamingApp
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp/streamingapp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp/streamingapp$ ls  -la
total 32
drwxr-xr-x 4 rajsaw rajsaw 4096 Mar  5 15:22 .
drwxrwxr-x 6 rajsaw rajsaw 4096 Mar  5 15:22 ..
drwxr-xr-x 2 rajsaw rajsaw 4096 Mar  5 15:22 charts
-rw-r--r-- 1 rajsaw rajsaw 1148 Mar  5 15:22 Chart.yaml
-rw-r--r-- 1 rajsaw rajsaw  349 Mar  5 15:22 .helmignore
drwxr-xr-x 3 rajsaw rajsaw 4096 Mar  5 15:22 templates
-rw-r--r-- 1 rajsaw rajsaw 5258 Mar  5 15:22 values.yaml
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp/streamingapp$ cd ..
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ code .
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ kubectl get svc -n streamingapp
NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
mongo                         ClusterIP   10.100.40.42    <none>        27017/TCP   34m
streamingapp-adminservice     ClusterIP   10.100.183.58   <none>        3003/TCP    80m
streamingapp-authservice      ClusterIP   10.100.8.254    <none>        3001/TCP    80m
streamingapp-chatservice      ClusterIP   10.100.85.215   <none>        3004/TCP    80m
streamingapp-frontend         ClusterIP   10.100.11.241   <none>        80/TCP      80m
streamingapp-streamingservice ClusterIP   10.100.151.7    <none>        3002/TCP    80m
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ kubectl edit svc streamingapp-frontend -n streamingapp

service/streamingapp-frontend edited
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ kubectl get svc -n streamingapp
NAME                          TYPE           CLUSTER-IP      EXTERNAL-IP                                                                PORT(S)        AGE
mongo                         ClusterIP      10.100.40.42    <none>                                                                     27017/TCP      37m
streamingapp-adminservice     ClusterIP      10.100.183.58   <none>                                                                     3003/TCP       84m
streamingapp-authservice      ClusterIP      10.100.8.254    <none>                                                                     3001/TCP       84m
streamingapp-chatservice      ClusterIP      10.100.85.215   <none>                                                                     3004/TCP       84m
streamingapp-frontend         LoadBalancer   10.100.11.241   ad21f81d5df8a414ea14ca82f7284dc0-1987682598.us-west-1.elb.amazonaws.com   80:32625/TCP   84m
streamingapp-streamingservice ClusterIP      10.100.151.7    <none>                                                                     3002/TCP       84m
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ kubectl get svc -n streamingapp
NAME                          TYPE           CLUSTER-IP      EXTERNAL-IP                                                                PORT(S)        AGE
mongo                         ClusterIP      10.100.40.42    <none>                                                                     27017/TCP      41m
streamingapp-adminservice     ClusterIP      10.100.183.58   <none>                                                                     3003/TCP       87m
streamingapp-authservice      ClusterIP      10.100.8.254    <none>                                                                     3001/TCP       87m
streamingapp-chatservice      ClusterIP      10.100.85.215   <none>                                                                     3004/TCP       87m
streamingapp-frontend         LoadBalancer   10.100.11.241   ad21f81d5df8a414ea14ca82f7284dc0-1987682598.us-west-1.elb.amazonaws.com   80:32625/TCP   87m
streamingapp-streamingservice ClusterIP      10.100.151.7    <none>                                                                     3002/TCP       87m
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~/var2/Assignmnets/StreamingApp$ docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED       STATUS                    PORTS                                      NAMES
28a1dfc81732   streamingapp-frontend  "/docker-entrypoint.…"   3 days ago    Exited (255) 2 days ago   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp    streamingapp-frontend-1
```

- To to change the streamingapp-frontend type from clusterIP to LoadBalancer I ran the command
- kubectl edit svc streamingapp-frontend -n streamingapp

Now I got the public url of frontend

http://ad21f81d5df8a414ea14ca82f7284dc0-1987682598.us-west-1.elb.amazonaws.com/

Looks I am done with deployment with helm

Current Architecture

GitHub
|
Webhook
|
Jenkins CI Pipeline

|

Docker Build

|

Push Images → AWS ECR

|

Helm Deploy

|

Amazon EKS

|

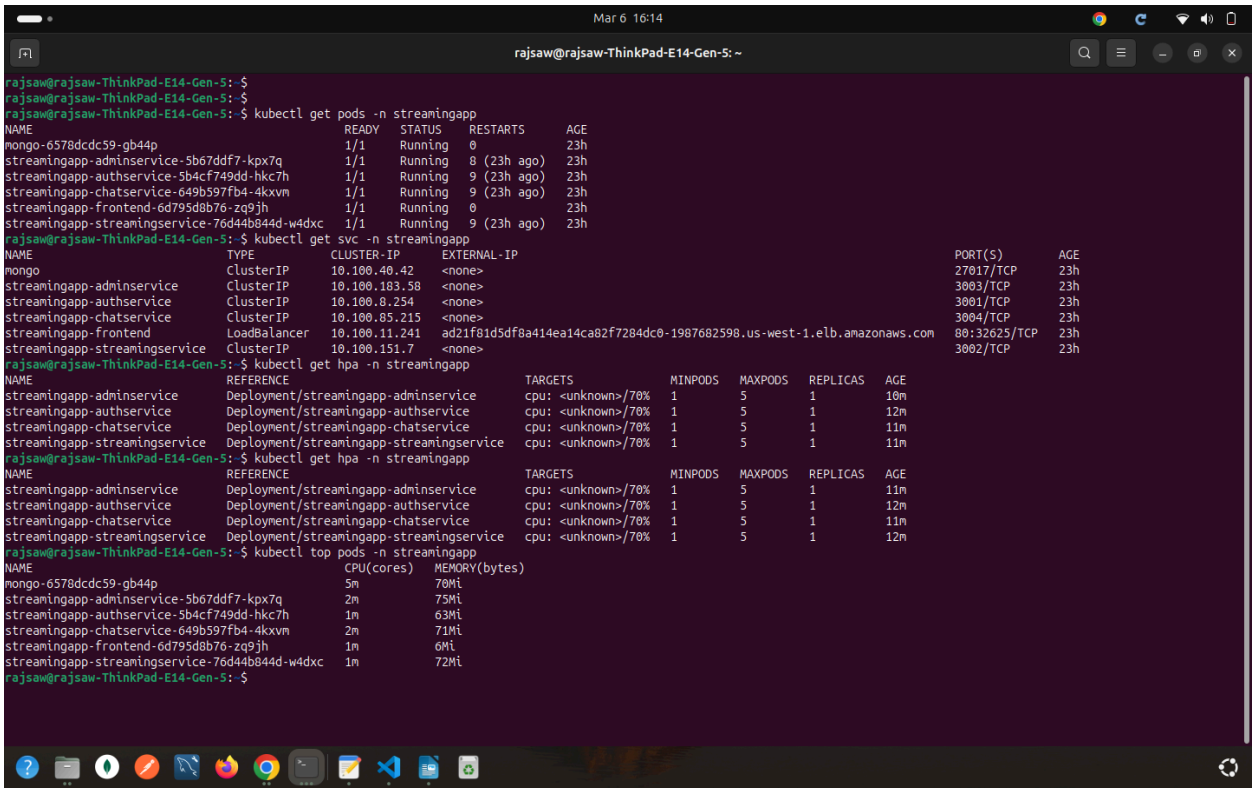
Pods Running

|

AWS LoadBalancer

|

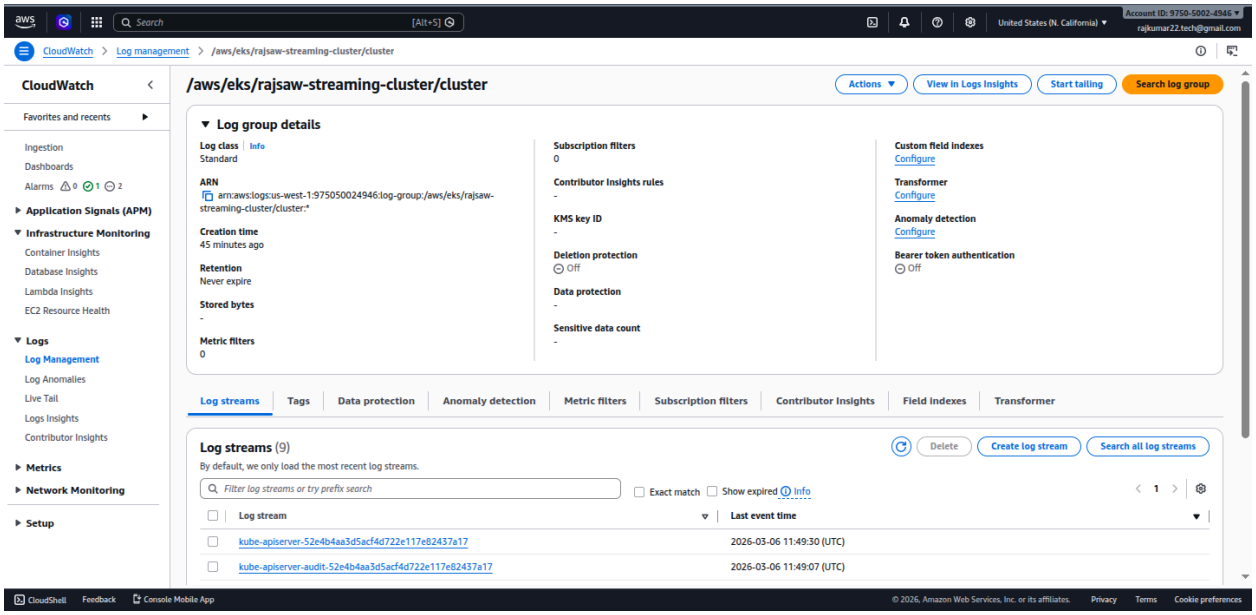
Public Access



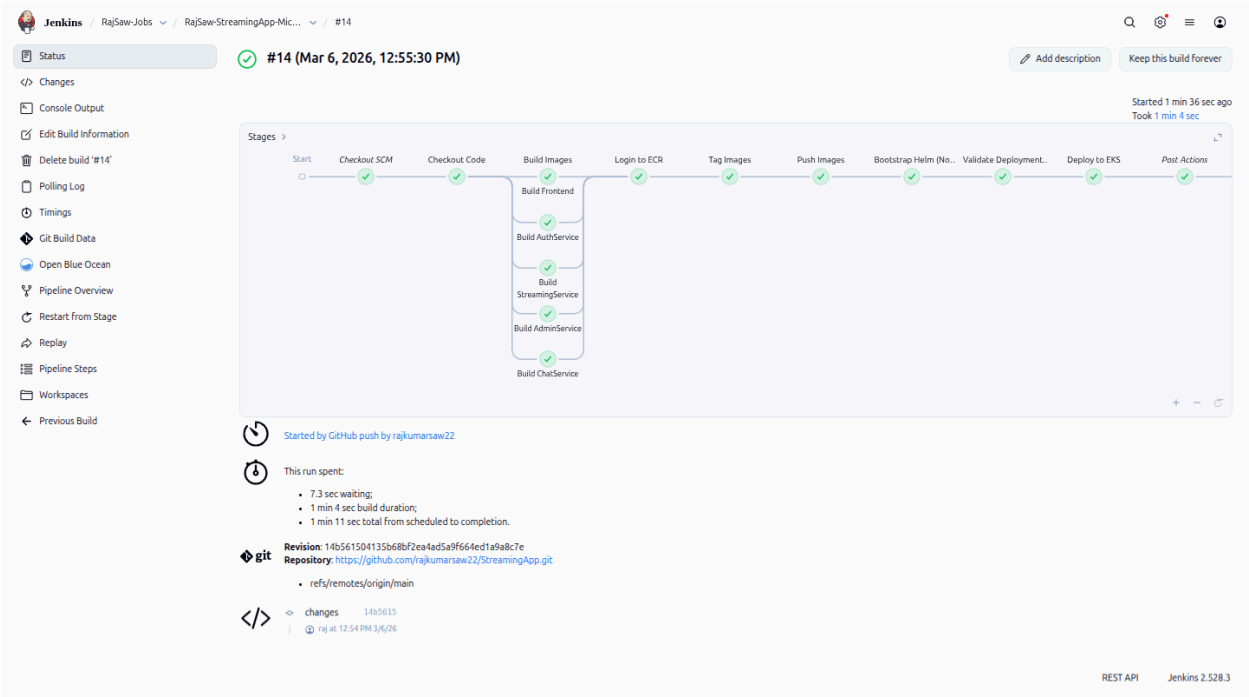
```
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ kubectl get pods -n amazon-cloudwatch
NAME                                                      READY   STATUS    RESTARTS   AGE
amazon-cloudwatch-observability-controller-manager-c8895f5hjhvp   1/1     Running   0          4m37s
cloudwatch-agent-6lr8x                                    1/1     Running   0          4m30s
cloudwatch-agent-7pgm9                                    1/1     Running   0          4m30s
fluent-bit-7vmdm                                          1/1     Running   0          4m38s
fluent-bit-8vq2c                                          1/1     Running   0          4m38s
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ kubectl get pods -n amazon-cloudwatch
NAME                                                      READY   STATUS    RESTARTS   AGE
amazon-cloudwatch-observability-controller-manager-c8895f5hjhvp   1/1     Running   0          12m
cloudwatch-agent-6lr8x                                    1/1     Running   0          12m
cloudwatch-agent-7pgm9                                    1/1     Running   0          12m
fluent-bit-7vmdm                                          1/1     Running   0          12m
fluent-bit-8vq2c                                          1/1     Running   0          12m
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ kubectl get daemonset -n amazon-cloudwatch
NAME                                  DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR              AGE
cloudwatch-agent                      2         2         2       2            2           kubernetes.io/os=linux     14m
cloudwatch-agent-windows              0         0         0       0            0           kubernetes.io/os=windows   14m
cloudwatch-agent-windows-container-insights   0         0         0       0            0           kubernetes.io/os=windows   14m
dcgm-exporter                         0         0         0       0            0           kubernetes.io/os=linux     14m
fluent-bit                            2         2         2       2            2           kubernetes.io/os=linux     14m
fluent-bit-windows                    0         0         0       0            0           kubernetes.io/os=windows   14m
neuron-monitor                        0         0         0       0            0           kubernetes.io/os=linux     14m
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ kubectl logs streamingapp-authservice-5b4cf749dd-hkc7h -n streamingapp

> authservice@1.0.0 start
> node index.js

Connecting to MongoDB at: mongodb://mongo:27017/streamingapp
(node:18) [MONGODB DRIVER] Warning: useNewUrlParser is a deprecated option: useNewUrlParser has no effect since Node.js Driver version 4.0.0 and will be removed in the next major
 version
(Use `node --trace-warnings ...` to show where the warning was created)
(node:18) [MONGODB DRIVER] Warning: useUnifiedTopology is a deprecated option: useUnifiedTopology has no effect since Node.js Driver version 4.0.0 and will be removed in the next
 major version
Authentication Service Started at port 3001
DB connection established
MongoDB Connected Successfully
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$ kubectl rollout restart deployment streamingapp-authservice -n streamingapp
deployment.apps/streamingapp-authservice restarted
rajsaw@rajsaw-ThinkPad-E14-Gen-5:~$
```



Final jenkins successful deployment screen



Source Code
GitHub
CI/CD Pipeline
Jenkins
• Build & Test
• Docker Images
Container Registry
Amazon ECR
• Amazon ECR
Kubernetes Cluster
Amazon EKS
• Pods & Services
• AWS Load Balancer
Monitoring
Amazon CloudWatch
• Logs & Metrics
• Container Insights